

Electric Potential, solved

Step-by-step solutions to every practice problem in Chapter 15. Try each problem on your own before you read its solution.

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P15.1

(a) No. Midway between two equal positive charges, the two fields cancel, so $\vec{E} = 0$. But both potentials are positive and add: $V = 2kq/(d/2) \neq 0$.

(b) No. At the midpoint of a dipole, the potentials of $+q$ and $-q$ cancel, so $V = 0$. But the two fields both point from $+$ towards $-$, so $\vec{E} \neq 0$.

The field depends on how quickly V changes, not on its value.

P15.2

Current flows through a body only if there is a potential **difference** across it. A bird's feet are both on the same wire, a few centimetres apart. The wire's resistance is so small that the potential difference between its feet is only millivolts, so a negligible current flows through the bird.

If the bird touches two different lines, or a line and a grounded support, it bridges a potential difference of thousands of volts. A large current then flows through it.

P15.3

Moving a charge along an equipotential involves no change in potential, so the field does no work: $\vec{E} \cdot d\vec{l} = 0$ for every small step $d\vec{l}$ within the surface. That means $\vec{E}$ is perpendicular to every direction lying in the surface, so it must be normal to the surface.

P15.4

$\Delta U = q\Delta V$ with $q = -e$ and $\Delta V > 0$, so $\Delta U < 0$. The potential energy **decreases**. With no other forces acting, energy conservation means the kinetic energy **increases**. Electrons naturally “fall” towards higher potential.

P15.5

Pairs. Four charges make $\binom{4}{2} = 6$ pairs: four sides of length a and two diagonals of length $a\sqrt{2}$.

With $\frac{kq^2}{a} = \frac{(8.99 \times 10^9)(1.0 \times 10^{-6})^2}{0.10} = 0.0899 \text{ J}$:

$$U = \frac{kq^2}{a} \left(4 + \frac{2}{\sqrt{2}} \right) = 0.0899(4 + 1.414) = 0.49 \text{ J}.$$

P15.6

Potential. Each charge is $a/\sqrt{2} = 0.0707 \text{ m}$ from the centre:

$$V = 4 \frac{kq}{a/\sqrt{2}} = \frac{4(8.99 \times 10^9)(1.0 \times 10^{-6})}{0.0707} = 5.1 \times 10^5 \text{ V}.$$

Field: $\vec{E} = 0$ by symmetry, since the four equal fields cancel in pairs.

The potential is large while the field is zero, another example for P15.1.

P15.7

Kinetic energy:

$$K = qV = (2e)(2.0 \times 10^6 \text{ V}) = 4.0 \text{ MeV} = 6.4 \times 10^{-13} \text{ J}.$$

Speed:

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(6.4 \times 10^{-13})}{6.64 \times 10^{-27}}} = 1.4 \times 10^7 \text{ m/s}.$$

That is about 4.6% of the speed of light, so the classical formula is fine.

P15.8

$$d = \frac{k(2e)(79e)}{K} = \frac{(8.99 \times 10^9)(158)(1.602 \times 10^{-19})^2}{7.7(1.602 \times 10^{-13})} = \frac{3.65 \times 10^{-26}}{1.23 \times 10^{-12}} = 3.0 \times 10^{-14} \text{ m} = 30 \text{ fm}.$$

This is still about 4 times the nuclear radius, so the alpha particle is turned back by the Coulomb force alone. Rutherford's scattering formula, which assumes a pure Coulomb force, therefore worked. Only at higher energies, or with lighter target nuclei, do deviations appear that reveal the nucleus's size and the strong nuclear force.

P15.9

Field:

$$E = \frac{\Delta V}{d} = \frac{500}{0.020} = 2.5 \times 10^4 \text{ V/m}.$$

Speed. The proton falls through 500 V, gaining $K = 500 \text{ eV}$:

$$v = \sqrt{\frac{2eV}{m_p}} = \sqrt{\frac{2(1.602 \times 10^{-19})(500)}{1.673 \times 10^{-27}}} = 3.1 \times 10^5 \text{ m/s}.$$

P15.10

Potentials. With $kQ = (8.99 \times 10^9)(1.0 \times 10^{-8}) = 89.9 \text{ V m}$:

$$V_{\text{centre}} = \frac{kQ}{R} = \frac{89.9}{0.10} = 899 \text{ V}, \quad V(x = R) = \frac{kQ}{\sqrt{2}R} = 636 \text{ V}.$$

Work to bring 1.0 nC from infinity (where $V = 0$) to the centre:

$$W = q\Delta V = (1.0 \times 10^{-9})(899) = 9.0 \times 10^{-7} \text{ J}.$$

At the centre the field is zero, but the potential is at its maximum along the axis, so energy is still needed to get there.

P15.11

$$E_x = -\frac{dV}{dx} = -(10x - 3) = 3 - 10x \text{ V/m.}$$

At $x = 2.0$ m: $E_x = 3 - 20 = -17$ V/m, so the field is 17 V/m in the $-x$ direction.

Equilibrium: $E_x = 0$ at $x = 0.30$ m.

V has a **minimum** there, since $d^2V/dx^2 = 10 > 0$. A positive charge, which seeks low potential, is in *stable* equilibrium along x at that point. A negative charge, which seeks high potential, would be unstable there.

P15.12

(a) **Field:**

$$E = \frac{\Delta V}{d} = \frac{0.070}{8.0 \times 10^{-9}} = 8.8 \times 10^6 \text{ V/m,}$$

directed inward, from the positive outside to the negative inside. That is roughly three times the breakdown field of air.

(b) **Energy.** The Na^+ ion moves from 0 V outside to -70 mV inside, down a potential “hill”, so it gains

$$\Delta K = e(70 \text{ mV}) = 0.070 \text{ eV} = 1.1 \times 10^{-20} \text{ J.}$$

The thermal energy is $kT = (1.38 \times 10^{-23})(310) = 4.3 \times 10^{-21} \text{ J} = 0.027 \text{ eV}$, so the energy gained is about 2.6 kT .

Significance. The membrane potential gives each sodium ion an energy comparable to a few times its thermal energy. That is enough to strongly bias the ions’ random motion inward when sodium channels open, but not so much that thermal fluctuations are irrelevant. The balance between electrical energy and thermal energy is what the Nernst equation of physiology describes.

P15.13

Rings. A ring of radius r and width dr carries $dq = \sigma 2\pi r dr$. Every part of it is a distance $\sqrt{x^2 + r^2}$ from the field point on the axis:

$$V = \int_0^R \frac{k\sigma 2\pi r dr}{\sqrt{x^2 + r^2}} = 2\pi k\sigma \left[\sqrt{x^2 + r^2} \right]_0^R = \frac{\sigma}{2\epsilon_0} \left(\sqrt{x^2 + R^2} - x \right).$$

Field:

$$E_x = -\frac{dV}{dx} = -\frac{\sigma}{2\epsilon_0} \left(\frac{x}{\sqrt{x^2 + R^2}} - 1 \right) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right). \checkmark$$

P15.14

(a) **Potential inside.** Outside the sphere, $V = kQ/r$, so at the surface $V(R) = kQ/R$. Inside, $E = kQr/R^3$, so

$$V(r) = V(R) - \int_R^r \frac{kQr'}{R^3} dr' = \frac{kQ}{R} - \frac{kQ}{2R^3} (r^2 - R^2) = \frac{kQ(3R^2 - r^2)}{2R^3}.$$

(b) At the centre:

$$V(0) = \frac{3kQ}{2R} = 1.5 V(R).$$

The potential keeps rising as you go inward, because there is still a field inside that a positive test charge would have to be pushed against.

P15.15

(a) **Building the sphere.** Suppose the sphere has been built up to radius r , with charge $q(r) = Q(r/R)^3$. Adding a thin shell of charge $dq = 3Qr^2 dr/R^3$ at its surface, where the potential is kq/r , takes work

$$dU = \frac{kq(r)}{r} dq = \frac{k}{r} \cdot \frac{Qr^3}{R^3} \cdot \frac{3Qr^2 dr}{R^3} = \frac{3kQ^2 r^4}{R^6} dr.$$

$$U = \int_0^R \frac{3kQ^2 r^4}{R^6} dr = \frac{3kQ^2}{5R}.$$

(b) **Uranium nucleus.** With $Q = 92e = 1.47 \times 10^{-17}$ C:

$$U = \frac{3(8.99 \times 10^9)(1.47 \times 10^{-17})^2}{5(7.4 \times 10^{-15})} = 1.6 \times 10^{-10} \text{ J} \approx 990 \text{ MeV}.$$

(c) **After fission.** Each fragment has charge $Q/2$ and radius $R/2^{1/3} = 0.794R$, so its self-energy is

$$U_{\text{frag}} = U \times \frac{(1/2)^2}{0.794} = 0.315 U.$$

The two fragments together have $0.63U$, so the electrostatic self-energy *falls* by about $0.37U \approx 370$ MeV.

Comparison. This is the right order of magnitude for the roughly 200 MeV released. Splitting the charge lowers the electrostatic repulsion energy, which is what drives fission. It is partly offset by an increase in *nuclear surface energy*: two smaller nuclei have more total surface, and the strong force binds surface nucleons less tightly. The net release is about 200 MeV, carried mostly as the kinetic energy of the fragments as they fly apart. Why heavy nuclei fission and light ones fuse is explored further in Chapter 24.

P15.16

(a) **Potential far away.** Put $+q$ at $z = +d/2$ and $-q$ at $z = -d/2$. For $r \gg d$, the distances to the two charges are approximately $r_{\pm} \approx r \mp \frac{d}{2} \cos \theta$. Then

$$V = kq \left(\frac{1}{r_+} - \frac{1}{r_-} \right) \approx kq \frac{r_- - r_+}{r^2} = \frac{kqd \cos \theta}{r^2} = \frac{kp \cos \theta}{r^2}.$$

(b) **Field components:**

$$E_r = -\frac{\partial V}{\partial r} = \frac{2kp \cos \theta}{r^3}, \quad E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{kp \sin \theta}{r^3}.$$

Checks against Chapter 13:

- On the axis ($\theta = 0$): $E_r = 2kp/r^3$ and $E_\theta = 0$. ✓
- On the equator ($\theta = 90^\circ$): $E_r = 0$ and $E_\theta = kp/r^3$. The direction of increasing θ at the equator is $-\hat{z}$, so the field is antiparallel to $\vec{p}$, with magnitude kp/r^3 . ✓

P15.17

(a) **Potential difference.** Between the conductors, $E = 2k\lambda/r$:

$$\Delta V = \int_a^b \frac{2k\lambda}{r} dr = 2k\lambda \ln \frac{b}{a}.$$

(b) Charge per metre for 1.0 kV:

$$\lambda = \frac{\Delta V}{2k \ln(b/a)} = \frac{1000}{2(8.99 \times 10^9) \ln 6} = \frac{1000}{3.22 \times 10^{10}} = 3.1 \times 10^{-8} \text{ C/m} = 31 \text{ nC/m}.$$

(c) Largest field, at the surface of the inner conductor:

$$E_{\max} = \frac{2k\lambda}{a} = \frac{\Delta V}{a \ln(b/a)} = \frac{1000}{(5.0 \times 10^{-4})(1.79)} = 1.1 \times 10^6 \text{ V/m}.$$

P15.18

(a) Kinetic energy: $K = 100 \text{ keV} = 1.0 \times 10^5 \times 1.602 \times 10^{-19} = 1.6 \times 10^{-14} \text{ J}.$

(b) Classical speed:

$$v = \sqrt{\frac{2K}{m_e}} = \sqrt{\frac{2(1.6 \times 10^{-14})}{9.11 \times 10^{-31}}} = 1.9 \times 10^8 \text{ m/s} = 0.63 c.$$

At such a large fraction of c , the classical formula can no longer be trusted, so we need the relativistic one.

(c) Relativistic speed.

$$\gamma = 1 + \frac{K}{m_e c^2} = 1 + \frac{100}{511} = 1.196.$$

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - 0.699} = 0.548 \Rightarrow v = 1.6 \times 10^8 \text{ m/s}.$$

The classical estimate is about 14% too high.

(d) Shortest wavelength (the Duane–Hunt limit):

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1240 \text{ eV nm}}{1.0 \times 10^5 \text{ eV}} = 0.0124 \text{ nm} = 12 \text{ pm}.$$

That is far shorter than an atom's diameter, which is why X-rays pass through soft tissue. Raising the tube voltage (“kVp” on a radiography console) produces more penetrating, shorter-wavelength X-rays.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are

listed in the References.

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